Reindexable RAG with Knowledge Graph

We present a RAG architecture that can be re-indexed when the embedding model changes, using cocoindex's hash-of-code invalidation. Cocoindex tracks each transformation by fingerprint of its source code; swapping the embed model rewrites the function body, invalidates the fingerprint, and triggers an automatic backfill.

A complementary knowledge graph is built by graphify using tree-sitter for code AST extraction and LLM semantic extraction for documents. The graph nodes carry confidence tags (EXTRACTED, INFERRED, AMBIGUOUS) so the LLM can weigh evidence.

Hybrid retrieval combines vector top-K from pgvector with graph neighbor expansion. This is shown to improve answer accuracy on entity-centric queries, and reduces hallucination by exposing structured citations to the LLM.